

Supplementary Section

Process Design, Techno-economic, and Life Cycle Assessment of Methanol Production Routes

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1. System Description

The specifications of every stream, component, and stream composition for each scenario employed for the methanol production are illustrated in the following tables.

Table S1: The specification of every stream in the BP-BCB scenario

Stream name	Temperature (°C)	Pressure (bar)	Mass flow (kg/h)
AFT-PYR	500.00	1.00	386000
B-AIR	25.00	1.00	550000
B-GAS	950.00	1.00	626000
BIOCHAR	30.00	1.00	46000
BIOGAS	25.00	1.00	76000
BIOMASS	25.00	1.00	386000
C-METH	40.00	1.00	208000
C-SYNGAS	30.00	1.00	446000
CB-GAS	30.00	1.00	626000
BIOCHAR1	500.00	1.00	46000
BIOCHAR-GAS	500.00	1.00	340000
CO2	35.00	1.00	120000
G-AIR	25.00	1.00	10000
G-AIR1	200.00	1.00	10000

Stream name	Temperature (°C)	Pressure (bar)	Mass flow (kg/h)
G-PRO	25.00	20.00	2000
G-PRO1	190.00	1.00	2000
G-PRO2	950.00	1.00	12000
G-PRO3	40.00	1.00	12000
H-BIO	500.00	1.00	386000
HCHAR-GAS	850.00	1.00	340000
KNO3	513.31	1.40	1.14×10^6
KNO3-1	930.00	1.40	1.14×10^6
KNO3-2	841.10	0.51	1.14×10^6
KNO3-3	801.13	0.51	1.72×10^6
KNO3-4	840.00	0.51	1.72×10^6
KNO3-5	576.40	0.00	1.72×10^6
KNO3-6	512.41	0.01	1.72×10^6
KNO3-7	512.73	0.51	1.72×10^6
KNO3-8	512.73	0.51	582000
KNO3-9	512.73	0.51	1.14×10^6
L-PRO	25.00	20.00	230000
M-GAS	25.00	1.00	352000
M-GAS1	35.00	30.00	233000
M-GAS2	145.7	44.00	233000
M-GAS3	105.279	44.00	233000
M-GAS4	157.4	65.00	233000
M-GAS5	196.00	65.00	233000
M-PRO1	194.00	65.00	233000
M-PRO2	25.00	65.00	233000
METH	64.47	1.00	208000
MIXTURE	25.00	20.00	229000
P	185.00	28.00	1.05×10^6
P1	96.24	0.68	1.05×10^6
P2	24.88	0.68	1.05×10^6
P3	30.27	28.00	1.05×10^6
R-AIR	25.00	1.00	71000
R-AIR1	400.00	1.00	71000
R-PRO	950.00	1.00	94000

Stream name	Temperature (°C)	Pressure (bar)	Mass flow (kg/h)
R-PRO1	40.00	1.00	94000
RESIDUAL	25.00	20.00	1000
ST1	25.00	1.00	106000
ST2	850.00	1.00	106000
SYNGAS	850.00	1.00	446000
T	400.00	60.00	529000
T1	236.72	0.04	529000
T2	25.09	0.04	529000
T3	36.61	60.00	529000
WA-METH	76.16	1.00	21000
WATER	25.00	1.00	93000

Table S2: The specification of every component employed in the BPBCB scenario

Heaters and Coolers			
Block Name	Pressure (bar)	Temperature (°C)	Heat Duty (kW)
C1	1.00	30	-8000
C2	1	30	-238000
C3	1.00	30	-284000
C4	44	105.279	-7100
C5	65	25	-46000
C6	1	40	-4000
C7	1	40	-5000
C8	1	40	-46000
COND1	0.01	512.40	-471000
COND2	0.04	25.09	-107000
COND3	0.68	24.87	-146000
H1	1.00	500	137000
H2	1.00	850	126000
<i>Continued on next page</i>			

Block Name	Pressure (bar)	Temperature (°C)	Heat Duty (kW)
H3	1.00	850	85000
H4	1	190	170
H5	1	200	504
H6	1	400	8000
H7	1.4	930	417000
H8	0.51	840	198000
H9	60	400	139000
H10	28	185	172000
H11	65	196	6800
Pumps			
Block Name	Specified Discharge Pressure (bar)	Power (kW)	Head Developed (m)
PUMP1	0.51	200	14.69
PUMP2	1.4	300	26.26
PUMP3	60	3400	711.68
PUMP4	28	4200	434.66
Compressors and Turbines			
Block Name	Specified Discharge Pressure (bar)	Isentropic Efficiency	Power (kW)
COMP1	44	0.85	8600
COMP2	65	0.85	9100
COMP _{PSA}	30	0.85	108000
TUR1	0.51	0.85	-25000
TUR2	0.01	0.85	-119000
TUR3	0.04	0.85	-36000
TUR4	0.68	0.85	-30000
Reactors			
<i>Continued on next page</i>			

Block Name	Pressure (bar)	Temperature (°C)	Heat Duty (kW)
Block Name	Outlet Temperature (°C)	Outlet Pressure (bar)	Heat Duty (kW)
B-BURNER	950	1	-241000
G-BURNER	950	1	-5000
R-BURNER	950	1	-30000
PYR	500	1	474000
GASIFIER	850	1	-224000
Methanol Reactor			
Block Name	Temperature (°C)	Pressure (bar)	Heat Duty (kW)
METH-R	194	65	-57081000
Distillation Tower			
Number of Stages	Molar Reflux Ratio	Condenser Temperature (°C)	Condenser Pressure (bar)
30	0.40	64.46	1
Condenser Duty (kW)	Reboiler Pressure (bar)	Reboiler Temperature (°C)	Reboiler Duty (kW)
-96000	1	76.15	105000

Table S3: The compositions of every stream in the BPBCB scenario

Stream	Composition (% mole)													
Stream	H2O	H2	O2	N2	CH4	CO	CO2	H2S	HCl	Methanol	Toluene	KNO3	N-Pentane	C
AFT-PYR	13.93	30.27	12.95	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.77
B-AIR	0.00	0.00	21.00	79.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
B-GAS	17.82	0.00	0.30	68.85	0.00	0.00	13.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BIOCHAR	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
BIOGAS	0.00	0.00	0.00	5.00	65.00	0.00	30.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CHAR	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
CHAR-GAS	15.74	34.22	14.64	0.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	35.30
CO2	0.00	0.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
G-AIR	0.00	0.00	21.00	79.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
G-PRO	0.02	36.28	0.00	14.68	3.47	42.49	1.95	0.04	0.21	0.86	0.00	0.00	0.00	0.00
G-PRO2	12.78	0.00	3.83	69.48	0.00	0.00	13.85	0.00	0.06	0.00	0.00	0.00	0.00	0.00
HCHAR-GAS	15.74	34.22	14.64	0.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	35.30
KNO3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00
L-PRO	8.34	0.05	0.00	0.06	0.04	0.08	0.24	0.03	0.05	91.12	0.00	0.00	0.00	0.00
M-GAS	2.69	57.03	0.00	0.09	0.03	28.78	11.35	0.01	0.02	0.00	0.00	0.00	0.00	0.00
M-PRO1	8.20	0.63	0.00	0.29	0.09	0.76	0.27	0.03	0.06	89.68	0.00	0.00	0.00	0.00
METH	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	99.00	0.00	0.00	0.00	0.00
MIXTURE	8.38	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	91.62	0.00	0.00	0.00	0.00
P	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00	0.00
R-AIR	0.00	0.00	21.00	79.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
R-PRO	34.66	0.10	0.15	54.66	0.08	0.17	10.02	0.00	0.11	0.00	0.00	0.00	0.00	0.00
RESIDUAL	0.00	8.70	0.00	10.39	6.73	15.10	44.21	4.63	10.10	0.00	0.00	0.00	0.00	0.00
ST1	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SYNGAS	19.99	46.89	0.00	0.08	0.02	23.67	9.34	0.01	0.01	0.00	0.00	0.00	0.00	0.00
T	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00	0.00
WA-METH	62.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	37.91	0.00	0.00	0.00	0.00
WATER	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table S4: The specifications of all the streams employed in the BWOB scenario

Stream name	Temperature (°C)	Pressure (bar)	Mass flow (kg/h)
AFT-PYR	500.00	1.00	386000
BIOMASS	25.00	1.00	386000
C-AIR	25.00	1.00	488000
BIOCHAR	500.00	1.00	46000
C-METH	40.00	1.00	208000
C-SYNGAS	30.00	1.00	447000
BIOCHAR-GAS	500.00	1.00	340000
CO2	35.00	1.00	120000
G-AIR	25.00	1.00	10000
<i>Continued on next page</i>			

Stream name	Temperature (°C)	Pressure (bar)	Mass flow (kg/h)
G-AIR1	200.00	1.00	10000
G-PRO	25.00	20.00	2000
G-PRO1	190.00	1.00	2000
G-PRO2	950.00	1.00	12000
G-PRO3	40.00	1.00	12000
GAS	950.00	1.00	534000
GAS1	30.00	1.00	534000
H-BIO	500.00	1.00	386000
HCHAR-GAS	850.00	1.00	340000
KNO3	513.31	1.40	1.12×10^6
KNO3-1	940.00	1.40	1.12×10^6
KNO3-2	850.43	0.51	1.12×10^6
KNO3-3	801.13	0.51	1.56×10^6
KNO3-4	840.00	0.51	1.56×10^6
KNO3-5	651.40	0.04	1.56×10^6
KNO3-6	653.20	0.04	1.56×10^6
KNO3-7	575.30	0.01	1.56×10^6
KNO3-8	512.41	0.01	1.56×10^6
KNO3-9	512.73	0.51	1.56×10^6
KNO3-10	512.73	0.51	440000
KNO3-11	512.73	0.51	1.12×10^6
L-PRO	25.00	20.00	230000
M-GAS	25.00	1.00	353000
M-GAS1	35.00	30.00	232000
M-GAS2	145.7	44.00	232000
M-GAS3	105.279	44.00	232000
M-GAS4	157.4	65.00	232000
M-GAS5	194.00	65.00	232000
M-PRO1	195.00	65.00	232000
M-PRO2	25.00	65.00	232000
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Stream name	Temperature (°C)	Pressure (bar)	Mass flow (kg/h)
METH	64.47	1.00	208000
MIXTURE	25.00	20.00	229000
P	185.00	28.00	809000
P1	86.76	0.68	809000
P2	24.88	0.68	809000
P3	30.27	28.00	809000
R-AIR	25.00	1.00	71000
R-AIR1	400.00	1.00	71000
R-PRO	950.00	1.00	93000
R-PRO1	40.00	1.00	93000
RESIDUAL	25.00	20.00	1000
ST1	25.00	1.00	108000
ST2	850.00	1.00	108000
SYNGAS	850.00	1.00	447000
T	400.00	60.00	299000
T1	226.60	0.08	299000
T2	39.62	0.08	879000
T3	330.00	0.08	879000
T4	315.01	0.04	879000
T5	25.09	0.04	879000
T6	25.10	0.08	879000
T7	25.10	0.08	580000
T8	25.10	0.08	299000
T9	36.61	60.00	299000
WA-METH	76.15	1.00	21000
WATER	25.00	1.00	94000

Table S5: The specifications of every component used in the BWOBB scenario

Heaters and coolers			
Block name	Pressure (bar)	Temperature (°C)	Heat duty (kW)
C1	1	30	-153000
C2	1.00	30	-286000
C3	44	105.279	-7100
C4	65	25	-46000
C5	1	40	-4000
C6	1	40	-5000
C7	1	40	-46000
COND1	0.01	512.40	-426000
COND2	0.04	25.09	-216000
COND3	0.68	24.87	-107000
H1	1.00	500	137000
H2	1.00	850	85000
H3	1.00	850	128000
H5	1	190	166
H6	1	200	504
H7	1	400	8000
H8	1.4	940	412000
H9	0.51	840	147000
H10	0.04	653.2	703
H11	60	400	79000
H12	0.08	330	165000
H13	28	185	132000
H14	65	194	6800
Pumps			
Block name	Specified discharge pressure (bar)	Power (kW)	Head developed (m)
<i>Continued on next page</i>			

Block name	Pressure (bar)	Temperature (°C)	Heat duty (kW)
PUMP1	0.51	212	14.69
PUMP2	1.4	272	26.26
PUMP3	0.08	4	0.47
PUMP4	60	2000	711.21
PUMP5	28	3000	434.66
Compressors and turbines			
Block name	Specified discharge pressure (bar)	Isentropic efficiency	Power (kW)
COMP1	44	0.85	8600
COMP2	65	0.85	9100
COMP _{PSA}	30	0.85	108000
TUR1	0.51	0.85	-10000
TUR2	0.04	0.85	-21000
TUR3	0.01	0.85	-9000
TUR4	0.08	0.85	-31000
TUR5	0.04	0.85	-4000
TUR6	0.68	0.85	-15000
Reactors			
Block name	Outlet temperature (°C)	Outlet pressure (bar)	Heat duty (kW)
C-BURNER	950	1	-242000
G-BURNER	950	1	-5000
R-BURNER	950	1	-30000
PYR	500	1	474000
GASIFIER	850	1	-224000
Methanol reactor			
Block name	Temperature (°C)	Pressure (bar)	Heat duty (kW)
METH-R	195	65	-238000
<i>Continued on next page</i>			

Block name	Pressure (bar)	Temperature (°C)	Heat duty (kW)
Distillation tower			
Number of stages	Molar reflux ratio	Condenser temperature (°C)	Condenser pressure (bar)
30	0.44	64.46	1
Condenser duty (kW)	Reboiler pressure (bar)	Reboiler temperature (°C)	Reboiler heat duty (kW)
-96000	1	76.15	105000

Table S6: The compositions of every stream in the BWOBB scenario

Stream	Composition (% mole)													
	H2O	H2	O2	N2	CH4	CO	CO2	H2S	HCl	Methanol	Toluene	KNO3	N-Pentane	C
AFT-PYR	13.93	30.27	12.95	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.77
C-AIR	0.00	0.00	21.00	79.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BIOCHAR	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
CHAR	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
BIOCHAR-GAS	15.74	34.22	14.64	0.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	35.30
CO2	0.00	0.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
G-AIR	0.00	0.00	21.00	79.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
G-PRO	0.02	52.45	0.00	14.86	3.44	26.06	2.08	0.04	0.21	0.85	0.00	0.00	0.00	0.00
G-PRO1	0.02	52.45	0.00	14.86	3.44	26.06	2.08	0.04	0.21	0.85	0.00	0.00	0.00	0.00
G-PRO2	17.13	0.00	4.09	69.62	0.00	0.00	9.09	0.00	0.06	0.00	0.00	0.00	0.00	0.00
GAS	0.00	0.00	0.04	79.00	0.00	0.00	20.96	0.00	0.00	0.00	0.00	0.00	0.00	0.00
HCHAR-GAS	15.74	34.22	14.64	0.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	35.30
KNO3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00
L-PRO	8.32	0.07	0.00	0.06	0.04	0.05	0.26	0.03	0.05	91.12	0.00	0.00	0.00	0.00
M-GAS	2.69	57.06	0.00	0.09	0.03	28.67	11.43	0.01	0.02	0.00	0.00	0.00	0.00	0.00
M-GAS1	3.04	64.43	0.00	0.10	0.03	32.37	0.00	0.01	0.02	0.00	0.00	0.00	0.00	0.00
M-PRO1	8.19	0.89	0.00	0.29	0.09	0.46	0.29	0.03	0.06	89.70	0.00	0.00	0.00	0.00
METH	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	99.00	0.00	0.00	0.00	0.00
MIXTURE	8.37	0.00	0.00	0.00	0.00	0.00	0.00	0.00	91.63	0.00	0.00	0.00	0.00	0.00
P	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
R-AIR	0.00	0.00	21.00	79.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
R-PRO	34.65	0.14	0.14	54.66	0.08	0.11	10.06	0.00	0.11	0.00	0.00	0.00	0.00	0.00
RESIDUAL	0.00	12.40	0.00	10.38	6.60	9.15	46.77	4.57	9.98	0.00	0.00	0.00	0.00	0.00
ST1	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SYNGAS	20.17	46.81	0.00	0.08	0.02	23.52	9.38	0.01	0.01	0.00	0.00	0.00	0.00	0.00
T	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00	0.00
WA-METH	62.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	37.95	0.00	0.00	0.00	0.00
WATER	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table S7: The specifications of all the streams employed in the BWOBWB scenario

Stream name	Temperature (°C)	Pressure (bar)	Mass flow (kg/h)
AFT-PYR	500.00	1.00	288000
B-AIR	25.00	1.00	494000
B-GAS	950.00	1.00	563000
BIOGAS	25.00	1.00	69000
BIOMASS	25.00	1.00	288000
C-METH	40.00	1.00	208000
C-SYNGAS	30.00	1.00	473000
CB-GAS	30.00	1.00	563000
BIOCHAR-GAS	850.00	1.00	288000
CO2	35.00	1.00	132000
G-AIR	25.00	1.00	19000
G-AIR1	200.00	1.00	19000
G-PRO	25.00	20.00	1000
G-PRO1	190.00	1.00	1000
G-PRO2	950.00	1.00	20000
G-PRO3	40.00	1.00	20000
H-BIO	500.00	1.00	288000
KNO3	940.00	1.40	1.09×10^6
KNO3-1	850.43	0.51	1.09×10^6
KNO3-2	801.13	0.51	1.32×10^6
KNO3-3	840.00	0.51	1.32×10^6
KNO3-4	651.40	0.04	1.32×10^6
KNO3-5	652.00	0.04	1.32×10^6
KNO3-6	574.20	0.01	1.32×10^6
KNO3-7	512.41	0.01	1.32×10^6
KNO3-8	512.73	0.51	1.32×10^6
KNO3-9	512.73	0.51	230000
KNO3-10	512.73	0.51	1.09×10^6
<i>Continued on next page</i>			

Stream name	Temperature (°C)	Pressure (bar)	Mass flow (kg/h)
KNO3-11	513.31	1.40	1.09×10^6
L-PRO	25.00	20.00	230000
M-GAS	25.00	1.00	363000
M-GAS1	35.00	30.00	231000
M-GAS2	146.05	44.00	231000
M-GAS3	105.6	44.00	231000
M-GAS4	178.46	75.00	231000
M-GAS5	190.00	75.00	231000
M-PRO1	190.00	75.30	231000
M-PRO2	25.00	75.30	231000
METH	64.47	1.00	208000
MIXTURE	25.00	20.00	230000
P	220.00	38.00	672000
P1	113.38	0.68	672000
P2	24.88	0.68	672000
P3	32.23	38.00	672000
R-AIR	25.00	1.00	72000
R-AIR1	400.00	1.00	72000
R-PRO	950.00	1.00	94000
R-PRO1	40.00	1.00	94000
RESIDUAL	25.00	20.00	602
ST1	25.00	1.00	184000
ST2	850.00	1.00	184000
SYNGAS	850.00	1.00	473000
T	560.00	108.00	178000
T1	367.89	0.08	178000
T2	39.62	0.08	1.13×10^6
T3	380.00	0.08	1.13×10^6
T4	364.62	0.04	1.13×10^6
T5	25.09	0.04	1.13×10^6
<i>Continued on next page</i>			

Stream name	Temperature (°C)	Pressure (bar)	Mass flow (kg/h)
T6	25.10	0.08	1.13×10^6
T7	25.10	0.08	962000
T8	25.10	0.08	178000
T9	45.64	108.00	178000
WA-METH	76.37	1.00	21000
WATER	25.00	1.00	109000

Table S8: The specifications of every component in the BWOBWB scenario

Heaters and Coolers			
Block name	Pressure (bar)	Temperature (°C)	Heat duty (kW)
C1	1	30	-215000
C2	1.00	30	-304000
C3	44	105.6	-7100
C4	75.3	25	-45000
C5	1	40	-90006
C6	1	40	-5000
C7	1	40	-46000
COND1	0.01	512.40	-360000
COND2	0.04	25.09	-313000
COND3	0.68	24.87	-100000
H1	1.00	500	102000
H2	1.00	850	72000
H3	1.00	850	219000
H4	1	190	303
H5	1	200	942
H6	1	400	8000
H7	1.4	940	400000
<i>Continued on next page</i>			

Block name	Pressure (bar)	Temperature (°C)	Heat duty (kW)
H8	0.51	840	76000
H9	0.04	652	263
H10	108	560	66000
H11	0.08	380	275000
H12	38	220	122000
H13	75	190	2000
Pumps			
Block name	Specified discharge pressure (bar)	Power (kW)	Head developed (m)
PUMP1	0.51	179	14.69
PUMP2	1.4	265	26.26
PUMP3	0.08	5	0.47
PUMP4	108	2100	1280.94
PUMP5	38	4000	593.76
Compressors and Turbines			
Block name	Specified discharge pressure (bar)	Isentropic efficiency	Power (kW)
COMP1	44	0.85	8700
COMP2	75	0.85	12800
COMP _{PSA}	30	0.85	109000
TUR1	0.51	0.85	-24000
TUR2	0.04	0.85	-65000
TUR3	0.01	0.85	-27000
TUR4	0.08	0.85	-18000
TUR5	0.04	0.85	-11000
TUR6	0.68	0.85	-27000
Reactors			
<i>Continued on next page</i>			

Block name	Pressure (bar)	Temperature (°C)	Heat duty (kW)
Block name	Outlet temperature (°C)	Outlet pressure (bar)	Heat duty (kW)
B-BURNER	950	1	-222000
G-BURNER	950	1	-9000
R-BURNER	950	1	-29000
PYR	500	1	354000
GASIFIER	850	1	-79000
Methanol Reactor			
Block name	Temperature (°C)	Pressure (bar)	Heat duty (kW)
METH-R	190	75	-239000
Distillation Tower			
Number of stages	Molar reflux ratio	Condenser temperature (°C)	Condenser pressure (bar)
30	0.40	64.46	1
Condenser duty (kW)	Reboiler pressure (bar)	Reboiler temperature (°C)	Reboiler heat duty (kW)
-97000	1	76.37	106000

Table S9: The compositions of every stream in the BWOBWB scenario

Stream	Composition (% mole)													
	H2O	H2	O2	N2	CH4	CO	CO2	H2S	HCl	Methanol	Toluene	KNO3	N-Pentane	C
AFT-PYR	13.93	30.27	12.95	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.78
B-AIR	0.00	0.00	21.00	79.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
B-GAS	18.08	0.00	0.00	68.71	0.00	0.00	13.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BIOGAS	0.00	0.00	0.00	5.00	65.00	0.00	30.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BIOCHAR-GAS	13.93	30.27	12.95	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	42.78
CO2	0.00	0.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
G-AIR	0.00	0.00	21.00	79.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
G-PRO	0.02	88.19	0.00	6.55	1.95	1.45	0.85	0.03	0.15	0.81	0.00	0.00	0.00	0.00
G-PRO2	26.66	0.00	3.47	68.39	0.00	0.00	1.44	0.00	0.04	0.00	0.00	0.00	0.00	0.00
KNO3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00
L-PRO	8.61	0.11	0.00	0.02	0.02	0.00	0.11	0.02	0.04	91.06	0.00	0.00	0.00	0.00
M-GAS	2.69	56.80	0.00	0.07	0.02	28.06	12.34	0.01	0.01	0.00	0.00	0.00	0.00	0.00
M-GAS1	3.07	64.80	0.00	0.08	0.03	32.01	0.00	0.01	0.02	0.00	0.00	0.00	0.00	0.00
M-PRO1	8.36	2.66	0.00	0.21	0.08	0.04	0.13	0.02	0.04	88.45	0.00	0.00	0.00	0.00
METH	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	99.00	0.00	0.00	0.00	0.00	0.00
MIXTURE	8.64	0.00	0.00	0.00	0.00	0.00	0.00	0.00	91.36	0.00	0.00	0.00	0.00	0.00
P	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
R-AIR	0.00	0.00	21.00	79.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
R-PRO	34.88	0.23	0.36	54.76	0.04	0.01	9.64	0.00	0.08	0.00	0.00	0.00	0.00	0.00
RESIDUAL	0.00	34.82	0.00	7.68	6.30	0.86	32.58	5.68	12.07	0.00	0.00	0.00	0.00	0.00
ST1	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
SYNGAS	21.72	45.69	0.00	0.05	0.02	22.57	9.93	0.00	0.01	0.00	0.00	0.00	0.00	0.00
T	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00	0.00
WA-METH	62.96	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	37.04	0.00	0.00	0.00	0.00
WATER	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table S10: The specifications of all the streams employed in the NGM scenario

Stream Name	Temperature (°C)	Pressure (bar)	Mass Flows (kg/h)
C-METH	40.00	1.00	208000
CO2	25.00	1.00	54000
G-PRO	25.00	60.00	1968000
L-PRO	25.00	60.00	328000
M1	29.51	1.00	428000
M2	319.17	10.00	428000
M3	30.00	9.70	428000
M4	30.00	9.70	407000
M5	247.74	60.00	407000
M6	96.49	60.00	2296000
<i>Continued on next page</i>			

Stream Name	Temperature (°C)	Pressure (bar)	Mass Flows (kg/h)
M7	190.00	60.00	2296000
M-PRO	190.00	60.00	2296000
M-PRO1	45.00	60.00	2296000
METH	65.06	1.00	2296000
MIXTURE	25.00	60.00	328000
NATURAL GAS	25.00	1.00	122000
NG-WA	29.25	1.00	598000
NG-WA1	800.00	1.00	598000
R-AIR	25.00	1.00	699000
R-AIR1	55.00	1.00	699000
R-FEED	25.00	60.00	79000
R-PRO	950.00	1.00	778000
R-PRO1	30.00	1.00	778000
R-WATER1	30.00	1.00	224000
R-WATER2	30.00	9.70	21000
RE-PRO	800.00	1.00	598000
RE-PRO1	366.00	1.00	598000
RECYCLE	25.00	60.00	1889000
RESIDUAL	25.00	60.00	86
W-PRO	487.86	1.00	598000
W-PRO1	30.00	0.70	598000
W-PRO2	30.00	1.00	374000
WA	790.00	19.00	325000
WA1	269.75	0.37	325000
WA2	76.09	0.37	325000
WA3	77.11	19.00	325000
WA-METH	102.08	1.00	120000
WA-METH1	30.00	1.00	120000
WATER	35.00	1.00	476000

Table S11: The specification of every component employed in the NGM scenario

Heaters and coolers			
Block name	Pressure (bar)	Temperature (°C)	Heat duty (kW)
C1	1	366	-200000
C2	0.7	30	-355000
C3	9.7	30	-113000
C4	60	45	-501000
C5	1	40	-3000
C6	1	30	-10000
C7	1	30	-351000
COND	0.36	76.09	-252000
H1	1	800	635000
H2	60	190	324000
H3	1	55	6000
H4	19	790	351000
Pumps			
Block name	Specified discharge pressure (bar)	Power (kW)	Head developed (m)
PUMP	19	519	201.46
Compressors and turbines			
Block name	Specified discharge pressure (bar)	Isentropic efficiency	Power (kW)
COMP1	10	0.85	98000
COMP2	60	0.85	71000
TUR	0.36	0.85	-100000
Reactors			
Block name	Outlet temperature (°C)	Outlet pressure (bar)	Heat duty (kW)

R-BURNER	950	1	-450000
Methanol WGS and Reformer reactors			
Block name	Temperature (°C)	Pressure (bar)	Heat duty (kW)
METH-R	190	60	-81000
WGS	366	1	0
REFORMER	800	1	463000
Distillation tower			
Number of stages	Molar reflux ratio	Condenser temperature (°C)	Condenser pressure (bar)
30	3.5119	65.06	1
Condenser duty (kW)	Reboiler pressure (bar)	Reboiler temperature (°C)	Reboiler heat duty (kW)
-316000	1	102.08	330000

Table S12: The compositions of the streams used in the NGM scenario

Stream	Composition (% mole)							
	CH4	CO2	CO	H2	H2O	N2	Methanol	O2
CO2	0.00	100.00	0.00	0.00	0.00	0.00	0.00	0.00
G-PRO	0.21	0.49	40.87	55.74	0.04	2.43	0.22	0.00
L-PRO	0.00	0.03	0.03	0.03	50.41	0.00	49.50	0.00
M1	0.02	21.30	4.85	70.27	3.27	0.29	0.00	0.00
M4	0.03	21.93	4.99	72.37	0.38	0.30	0.00	0.00
M6	0.15	6.96	30.04	60.76	0.14	1.79	0.16	0.00
M-PRO	0.17	0.42	34.61	47.20	7.76	2.06	7.78	0.00
METH	0.00	0.00	0.00	0.00	0.10	0.00	99.90	0.00
MIXTURE	0.00	0.00	0.00	0.00	50.46	0.00	49.54	0.00
NG	97.96	0.00	0.00	0.53	0.00	1.51	0.00	0.00
<i>Continued on next page</i>								

Stream	CH4	CO2	CO	H2	H2O	N2	Methanol	O2
NG-WA	20.62	0.00	0.00	0.11	78.95	0.32	0.00	0.00
R-AIR	0.00	0.00	0.00	0.00	0.00	79.00	0.00	21.00
R-FEED	0.21	0.49	40.87	55.74	0.04	2.43	0.22	0.00
R-PRO	0.00	7.32	0.00	0.00	9.91	72.26	0.00	10.51
R-WATER1	0.00	0.00	0.00	0.00	100.00	0.00	0.00	0.00
RE-PRO	0.02	0.00	14.59	43.84	41.33	0.23	0.00	0.00
RECYCLE	0.21	0.49	40.87	55.74	0.04	2.43	0.22	0.00
RESIDUAL	1.22	32.44	34.82	27.31	0.00	4.22	0.00	0.00
W-PRO	0.02	10.82	3.77	54.66	30.51	0.23	0.00	0.00
W-PRO2	0.03	15.02	5.23	75.88	3.53	0.31	0.00	0.00
WA	0.00	0.00	0.00	0.00	100.00	0.00	0.00	0.00
WA-METH	0.00	0.00	0.00	0.00	100.00	0.00	0.00	0.00
WATER	0.00	0.00	0.00	0.00	100.00	0.00	0.00	0.00

Table S13: The specifications of every stream used in the DCM scenario

Stream Name	Temperature (°C)	Pressure (bar)	Mass Flows (kg/h)
AIR	25.00	1.00	16000
C-FEED	35.00	1.20	23000
C-METH	35.00	1.00	208000
C-PRO	403.64	50.00	23000
C-WATER	35.00	1.00	71000
CO2	25.00	1.00	254800
CO2-1	222.65	7.00	255000
CO2-2	35.00	7.00	255000
CO2-3	131.85	19.00	255000
CO2-4	35.00	19.00	255000
CO2-5	130.40	50.00	255000
G-PRO	35.00	50.00	148000
<i>Continued on next page</i>			

Stream Name	Temperature (°C)	Pressure (bar)	Mass Flows (kg/h)
G-PRO1	35.00	1.20	24000
H2	25.00	1.00	35000
H2-1	308.88	8.00	35000
H2-2	35.00	8.00	35000
H2-3	136.98	19.00	35000
H2-4	35.00	19.00	35000
H2-5	151.02	50.00	35000
L-PRO	35.00	50.00	305000
M-PRO	195.00	50.00	453000
M-PRO1	35.00	50.00	453000
METH	64.32	1.00	208000
MIX	141.34	50.00	289000
MIX-1	123.18	50.00	453000
MIX-2	195.00	50.00	453000
MIXTURE	35.00	1.20	281000
P	185.00	27.00	2.69×10^6
P1	138.99	7.38	2.69×10^6
P2	110.03	7.38	2.69×10^6
P3	114.77	27.00	2.69×10^6
R-FEED	35.00	50.00	7000
R-FFED1	35.00	1.20	1000
R-PRO	950.00	1.00	24000
R-PRO1	35.00	1.00	24000
RECYCLE	35.00	50.00	141000
RESIDUAL	35.00	1.20	1000
WA-METH	35.00	1.20	280000
WATER	97.14	1.00	71000

Table S14: The specifications of every component employed in the DCM scenario

Heaters and coolers			
Block name	Pressure (bar)	Temperature (°C)	Heat duty (kW)
C1	8	35	-38000
C2	19	35	-14000
C3	7	35	-13000
C4	19	35	-7000
C5	50	35	-146000
C6	1	35	-0.54
C7	1	35	-0.33
C8	1	35	-0.30
COND	7.38	110.02	-263000
H	50	195	21000
H1	27	185	284000
Pumps			
Block name	Specified discharge pressure (bar)	Power (kW)	Head developed (m)
PUMP	27	9000	379.81
Compressors and turbines			
Block name	Specified discharge pressure (bar)	Isentropic efficiency	Power (kW)
COMP1	8	0.85	2.17
COMP2	19	0.85	0.77
COMP3	50	0.85	0.89
COMP4	7	0.85	0.74
COMP5	19	0.85	0.34
COMP6	50	0.85	0.32
COMP7	50	0.85	0.13
TUR	7.38	0.85	-29000
Reactors			

Block name	Outlet temperature (°C)	Outlet pressure (bar)	Heat duty (kW)
R-BURNER	950	1	-0.26
Methanol reactor			
Block name	Temperature (°C)	Pressure (bar)	Heat duty (kW)
METH-R	195	50	-142784.4
Distillation tower			
Number of stages	Molar reflux ratio	Condenser temperature (°C)	Condenser pressure (bar)
30	1.05	64.32	1
Condenser duty (kW)	Reboiler pressure (bar)	Reboiler temperature (°C)	Reboiler heat duty (kW)
-140000	1	97.14	151000

Table S15: The compositions of the streams in the DCM scenario

Stream	Composition (% mole)								
	H2O	H2	O2	N2	CH4	CO	CO2	Methanol	N-Pentane
AIR	0.00	0.00	21.00	79.00	0.00	0.00	0.00	0.00	0.00
C-FEED	2.15	1.82	0.00	0.00	0.00	0.00	81.30	14.73	0.00
CO2	0.00	0.00	0.00	0.00	0.00	40.24	59.76	0.00	0.00
G-PRO	0.08	54.06	0.00	0.00	0.00	0.10	45.19	0.57	0.00
G-PRO1	2.15	1.82	0.00	0.00	0.00	0.00	81.30	14.73	0.00
H2	0.00	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
L-PRO	34.97	0.10	0.00	0.00	0.00	0.00	4.59	60.34	0.00
M-PRO	21.45	21.01	0.00	0.00	0.00	0.04	20.32	37.18	0.00
METH	0.10	0.00	0.00	0.00	0.00	0.00	0.00	99.90	0.00
MIX	0.00	71.69	0.00	0.00	0.00	11.39	16.92	0.00	0.00
MIX-1	0.06	66.69	0.00	0.00	0.00	8.78	24.09	0.39	0.00
MIXTURE	36.84	0.00	0.00	0.00	0.00	0.00	0.22	62.94	0.00
P	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
<i>Continued on next page</i>									

Stream	H2O	H2	O2	N2	CH4	CO	CO2	Methanol	N-Pentane
R-FEED	0.08	54.06	0.00	0.00	0.00	0.10	45.19	0.57	0.00
R-PRO	22.76	0.00	2.38	51.62	0.00	0.00	22.37	0.87	0.00
RECYCLE	0.08	54.06	0.00	0.00	0.00	0.10	45.19	0.57	0.00
RESIDUAL	0.00	0.03	0.00	0.00	0.00	0.00	99.97	0.00	0.00
WA-METH	36.92	0.00	0.00	0.00	0.00	0.00	0.00	63.08	0.00
WATER	98.31	0.00	0.00	0.00	0.00	0.00	0.00	1.68	0.00

2. Life Cycle Assessment

The process trees of different scenarios are illustrated in Figs. S1 to S4.

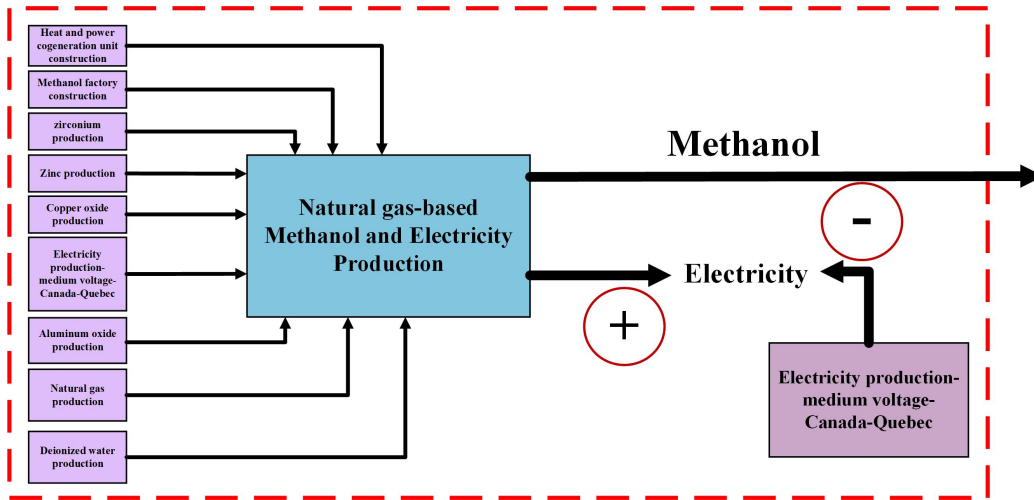


Figure S1: The process tree and system boundary of the NGM scenario

3. Economic Analysis

This section provides a detailed explanation of the methodologies employed for the economic analysis of the proposed methanol production scenarios. It includes descriptions of the approaches used for estimating capital costs, manufacturing costs, and conducting profitability analysis.

3.1. Capital Cost

The grassroot cost (C_{GR}), known as one of the module costing techniques presented by Turton et al. [1], is employed to calculate the capital cost of

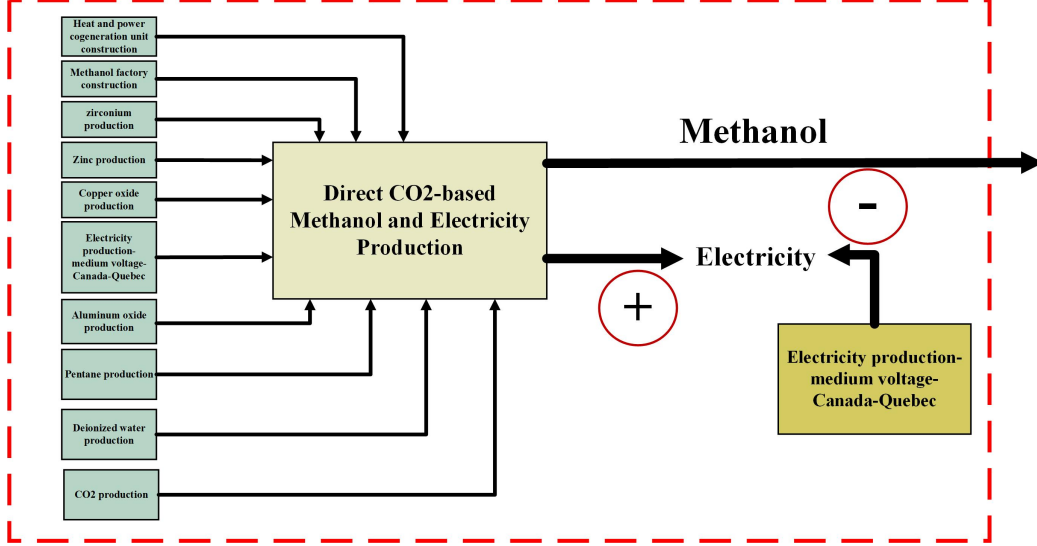


Figure S2: The process tree and system boundary of the DCM scenario

the plant. This term refers to the cost of new equipment needed for starting construction on undeveloped land. The equation used for calculating the purchased cost of each piece of equipment, based on the assumption of carbon steel construction and ambient operating pressure, is given as follows:

$$\log_{10}(C_{P_i}^0) = k_{1i} + k_{2i} \log_{10}(S_i) + k_{3i} (\log_{10}(S_i))^2 \quad (S1)$$

Where k_1 , k_2 , and k_3 are equipment factors, and S is the hardware size, such as the area of heat exchangers and the volume of towers and reactors.

The chemical engineering plant cost index (CEPCI) is employed to account for inflation rate between the reference and the base year in order to actualize the purchased cost of equipment, which can be acquired using the following equation:

$$C_{P_i} = C_{P_i}^0 \frac{\text{CEPCI}_{2023}}{\text{CEPCI}_{\text{ref}}} \quad (S2)$$

Then the bare module (BM) of every component can be calculated as follows:

$$C_{B_i} = C_{P_i} (b_{1i} + b_{2i} f_{p_i} f_{m_i}) \quad (S3)$$

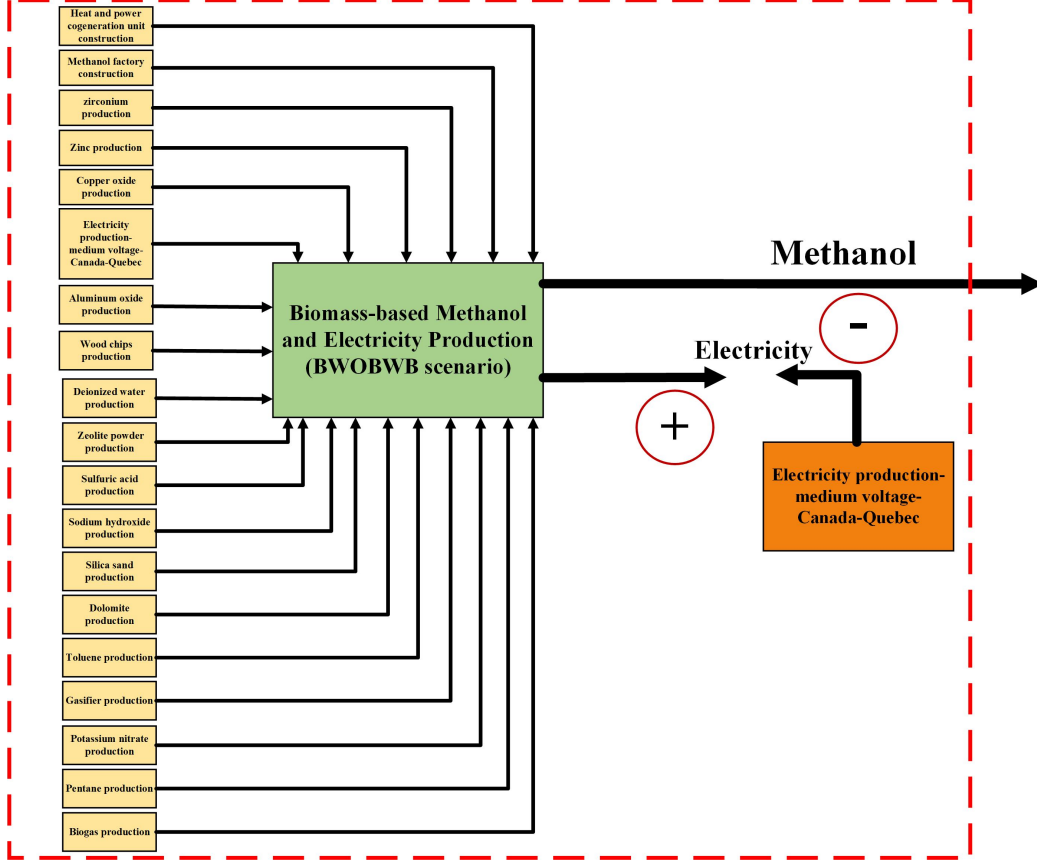


Figure S3: The process tree and system boundary of the BWOBWB scenario

Where b_1 and b_2 are equipment-based factors, and f_p and f_m are factors that account for effective operating pressure and construction material. The bare module costs can be rewritten for conditions where there is no need for special materials or to account for specific operating pressure conditions as follows:

$$C_{BM_{i0}} = C_{P_i}(b_{1i} + b_{2i}) \quad (\text{S4})$$

The C_{GR} can be calculated using the following equation.

$$C_{GR} = (1 + \alpha_1 + \alpha_2) \sum_{i=1}^n C_{BM_i} + \alpha_3 \sum_{i=1}^n C_{BM_i}^0 \quad (\text{S5})$$

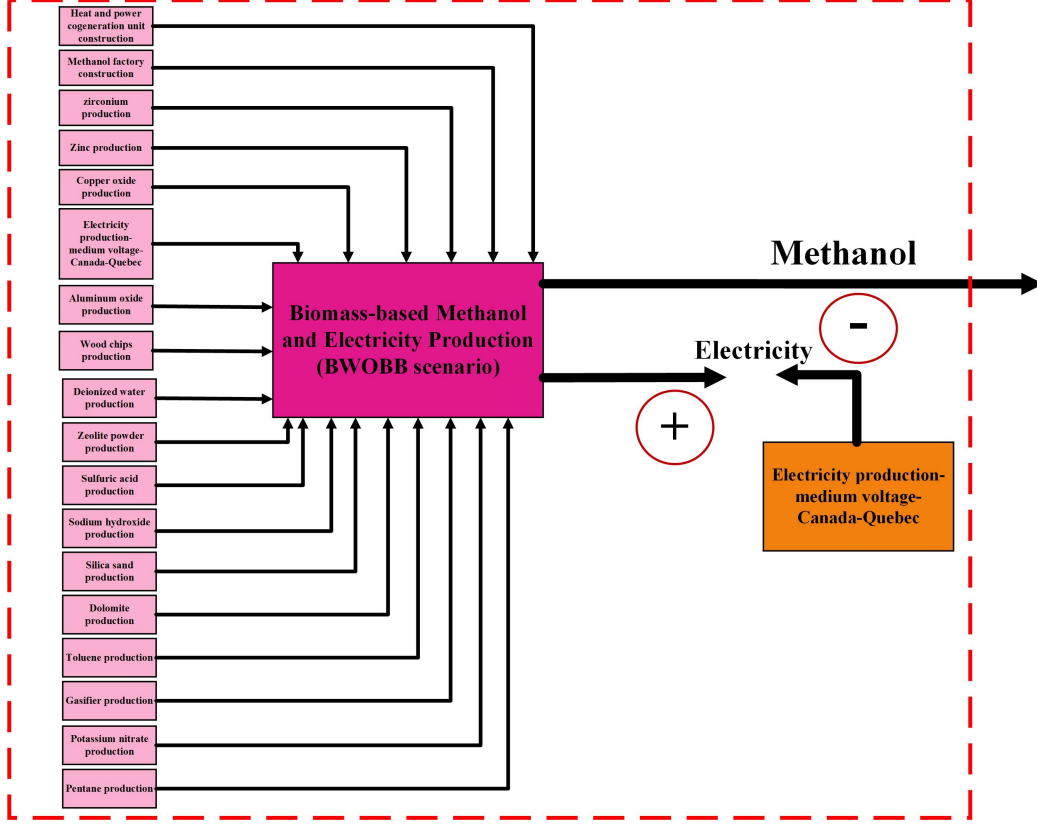


Figure S4: The process tree and system boundary of the BWOBB scenario

Where α_1 with the value of 0.35 is for flexible and electricity storage. In this equation, α_2 with the value of 0.03 is for fees. Also, α_3 with the value of 0.5 is for auxiliary facilities costs [1, 2]. Other empirical laws can be employed if there are no available cost correlations for the required equipment in [3, 1]. Therefore, when the costs based on the desired component size are not determined, they should be scaled using the power law, as shown below.

$$C_{BM_i} = C_{BM_i, \text{ref}} \left(\frac{S_i}{S_{i, \text{ref}}} \right)^\mu \quad (\text{S6})$$

Where $C_{BM, \text{ref}}$ is the reference bare module cost, S_{ref} is the reference size, and μ is the scaling factor. Bare module costs based on the reference size for various components can be obtained from [3, 2].

The capital cost of the electrolyzer used in the DCM scenario is calculated

based on the electricity required for producing hydrogen, which is equal to 39.4 kWh/kg_{H2}. Therefore, by considering the price of the electrolyzer based on power consumption, which is equal to 600 \$/kW, its capital cost can be determined [4, 5].

3.2. Manufacturing Cost

In this section, the approach explained by Turton et al. [1] is employed for calculating the manufacturing cost (C_M), including direct and indirect costs. Direct manufacturing costs account for raw materials, catalysts, and utilities, such as electricity and cooling water, which depend on market conditions, plant locations, and policies. The price of fuel used in each scenario for methanol production is illustrated in Table 11. On the other hand, the indirect costs comprise maintenance, labor, insurance, and taxes. In this study, it is assumed that the proposed plant is located in Quebec.

3.3. Profitability Analysis

After calculating the capital expenditures (CapEx) and operating expenditures (OpEx), the next step in assessing the economic viability of the methanol production processes is to conduct a profitability analysis. This analysis involves calculating three key financial metrics: the discounted payback period (DPP), the discounted net present value (NPV), and the discounted cash flow rate of return (DCFRROR).

The DPP is determined by calculating the present value (PV) of each year's cash flows using a discount rate. These discounted cash flows are cumulatively summed until they equal the initial CapEx. The year in which this sum first matches or exceeds the initial investment is identified as the payback period. This metric indicates how long it will take to recover the initial investment in today's dollars [1].

The NPV is calculated by discounting all future cash inflows and outflows to their present value using a chosen discount rate. The NPV is obtained by subtracting the present value of cash outflows, including CapEx and OpEx, from the present value of cash inflows over the project's lifetime. A positive NPV indicates that the project is expected to generate more value than it costs, making it financially viable.

The DCFRROR is found by determining the discount rate that makes the NPV of the project equal to zero. This is typically done through iterative methods using financial calculators or software. The DCFRROR represents the project's expected annualized return on investment, and if it exceeds the

project's cost of capital, it indicates a profitable venture [1]. The results of the CapEx, OpEx, and profitability can be found in Tables S16 and S17.

Table S16: The results of CapEx calculations

Equipment	Purchased Equipment Cost (\$)	Bare Module Cost (\$)	Total Grass Roots Cost (\$)
NGM scenario			
Compressors	68,300,000	187,000,000	-
Drives	508,000	760,000	-
Heat Exchangers	138,500,000	834,000,000	-
Pumps	86,900	600,000	-
Reactors	92,500,000	370,500,000	-
Towers	26,800,000	42,400,000	-
Turbines	31,600,000	110,400,000	-
CO2 Capture	94,600,000	94,600,000	-
Total	452,900,000	1,640,400,000	2,284,890,000
BPBCB scenario			
Compressors	42,700,000	117,100,000	-
Drives	11,590,000	17,370,000	-
Heat Exchangers	101,719,000	406,986,000	-
Pumps	897,000	11,682,000	-
Reactors	45,300,000	181,200,000	-
Towers	9,200,000	14,600,000	-
Turbines	127,300,000	446,000,000	-
Gasification Process	514,200,000	514,200,000	-
Total	853,000,000	1,709,200,000	2,478,700,000
BWOBB scenario			
Compressors	42,700,000	117,100,000	-
<i>Continued on next page</i>			

Drives	11,500,000	17,300,000	-
Heat Exchangers	101,700,000	406,900,000	-
Pumps	897,000	11,600,000	-
Reactors	45,300,000	181,200,000	-
Towers	9,200,000	14,600,000	-
Turbines	127,300,000	446,000,000	-
Gasification Process	514,200,000	514,200,000	-
Total	853,000,000	1,709,200,000	2,006,400,000
BWOBWB scenario			
Compressors	45,100,000	123,400,000	-
Drives	12,400,000	18,500,000	-
Heat Exchangers	155,100,000	504,200,000	-
Pumps	955,000	9,700,000	-
Reactors	45,300,000	181,000,000	-
Towers	9,300,000	14,700,000	-
Turbines	70,300,000	246,400,000	-
Gasification Process	435,300,000	435,300,000	-
Total	773,900,000	1,533,500,000	2,309,000,000
DCM scenario			
Compressors	40,700,000	111,600,000	-
Drives	11,600,000	17,500,000	-
Heat Exchangers	34,300,000	90,100,000	-
Pumps	130,000	9,100,000	-
Reactors	44,400,000	177,500,000	-
Towers	13,600,000	21,500,000	-
Turbines	8,100,000	28,400,000	-
Electrolyser	817,800,000	817,800,000	-
<i>Continued on next page</i>			

CO2 Capture	94,600,000	94,600,00	-
Total	153,000,000	455,800,000	1,821,200,000

Table S17: The results of OpEx and profitability calculations

Parameters	NGM scenario	BPBCB scenario	BWOBB scenario	BWOBBW scenario	DCM scenario
OpEx					
CRM (Raw Materials Costs) (\$)	164,227,919	477,100,000	261,700,000	324,000,000	27,300,000
CUT (Cost of Utilities) (\$)	160,000	39,000,000	77,200,000	41,200,000	588,000,000
Cost of Manufacturing (\$)	553,700,000	1,001,900,000	737,700,000	779,000,000	1,050,900,000
Total Module Factor	1.18	1.18	1.18	1.18	1.18
Grass Roots Factor	0.5	0.5	0.5	0.5	0.5
COL (Cost of Operating Labor) (\$)	1,000,000	1,400,000	1,000,000	1,400,000	1,200,000
Revenue From Sales (\$)	1,213,600,000	1,568,700,000	1,213,600,000	1,213,600,000	1,213,600,000
Construction Period (year)	3	3	3	3	3
Discounted Profitability Criterion					
Net Present Value (M\$)	2,714.48	2,043.91	1,667.77	1,369.08	-194.26
Discounted Payback Period (years)	4.9	6.2	6.5	7.4	-
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Discounted Cash Flow Rate of Return (%)	20.24	17	16.45	14.79	5.46
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